

Investment Portfolio Analysis

IRR and TWR Pipeline

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1 Introduction

This project presents a deterministic and auditable framework for investment portfolio performance analysis using two commonly used methodologies:

- money-weighted return (IRR)
- time-weighted return (TWR)

The implementation is based on real data exported from a broker and is designed with a strong emphasis on reproducibility, transparency, and modular architecture.

A key motivation of this work is the observation that commonly available portfolio analysis often lacks transparent methodological documentation and consistency of reported values.

2 Data Sources

The system operates with the following input data:

- `a_pokyny_realizovane.xlsx` – executed transactions
- `b_portfolio.xlsx` – current portfolio overview
- `c_cash_flow.xlsx` – transaction history
- `fx.<year>.txt` – daily exchange rates, source: Czech National Bank

All monetary values are normalized to CZK.

2.1 Data Quality Considerations

Several structural issues were identified during the analysis:

- **Absence of a consistent unique identifier across data sources**

Traded securities are not consistently identified by a single key across all files. While ISIN is available in transaction data, the current portfolio overview uses market-specific identifiers or naming conventions, which makes direct linkage non-trivial.

- **Ambiguity in the identification of traded securities**

The same security may appear under different market listings (e.g. on different exchanges or in different currencies), without the existence of a unique mapping key. This requires manual disambiguation when reconstructing positions and valuations.

- **Incomplete linkage between transactions and portfolio positions**

Executed transactions and portfolio items cannot be matched directly row by row. For example, the value of a position reported in the current portfolio overview cannot be traced back to a unique sequence of transactions without explicit reconstruction of portfolio evolution.

- **Presence of derived values without transparent construction logic**

Some reported quantities (e.g. total portfolio value or performance indicators) are provided without sufficient detail needed to reconstruct them from the underlying data. This makes independent verification of reported metrics impossible.

3 Architecture

The project follows a modular pipeline structure:

- **IO layer** – loading and normalization of source data
- **Transformation layer** – exchange-rate conversion and aggregation
- **Core computation** – IRR and TWR calculations
- **Audit layer** – intermediate tables for verification

All transformations are explicit, reproducible, and free of implicit state or hidden dependencies.

4 Branch 1: Money-Weighted Return (IRR)

4.1 Methodology

This branch is based on the standard concept of money-weighted return formulated as an equality of discounted, irregularly spaced external cash flows.

The internal rate of return, IRR, is denoted by r and is implicitly defined by the equation:

$$\sum_{k=1}^n \frac{CF_k}{(1+r)^{\tau_k}} + \frac{FV_d}{(1+r)^{\tau_d}} = 0 \quad (1)$$

where:

- CF_k denotes the k -th external cash flow included in the analysis
- d denotes the evaluation date
- FV_d denotes the final portfolio value at the evaluation date d
- τ_k is the length of the time interval (in years) between the reference date and the realization date of the k -th external cash flow
- τ_d is the length of the time interval (in years) between the reference date and the evaluation date d

In the implementation, the reference date is chosen as the realization date of the first external cash flow CF_1 from the transaction history. All external cash flows, as well as the final portfolio value, are discounted to this reference date.

In line with common conventions, cash inflows are treated as negative cash flows, while cash outflows and the final portfolio value are treated as positive cash flows.

The rate r is obtained from the transcendental equation (1) numerically using the bisection method.

4.2 Definition of External Cash Flows

External cash flows include:

- cash deposit
- cash withdrawal
- fee refund

4.3 Output Metrics

In addition to IRR, the outputs also include an auxiliary metric zisk_d defined as

$$\text{zisk}_d = \frac{FV_d}{\sum CF_k^*} - 1,$$

where CF_k^* are those external cash flows up to the evaluation date d that represent cash deposits.

This metric provides a simple relative expression of total appreciation without accounting for the time structure of external cash flows. However, this interpretation is valid only if no other external cash flows occur in the given period.

5 Branch 2: Time-Weighted Return (TWR)

5.1 Important Limitation

Due to the absence of real historical prices, the current implementation operates in **snapshot mode**.

5.2 Portfolio Value

The total portfolio value at time t is

$$V_t = \text{positions}_t + \text{cash}_t,$$

Portfolio positions (positions_t) and cash balance (cash_t) are reconstructed from transaction history.

5.3 Methodology

Time-weighted return is constructed as a sequence of period returns that isolate portfolio performance from external cash flows.

For the k -th external cash flow CF_k , let t_k denote the time of its realization. Then

$$r_k = \frac{V_{t_k} - V_{t_{k-1}} - CF_k}{V_{t_{k-1}}}, \quad k = 2, 3, \dots$$

is the portfolio return over the period from t_{k-1} to t_k ; if the length of this period is one day, the term daily portfolio return is used.

Subtracting CF_k removes the effect of external capital movements, so r_k reflects only the performance of invested assets.

The overall time-weighted return is then obtained by chaining the individual returns r_k :

$$TWR = \prod_k (1 + r_k) - 1 \quad (2)$$

This multiplicative structure ensures that each period contributes to the overall result proportionally, independently of the size and timing of external cash flows.

In a fully consistent setting with daily portfolio returns, this formulation provides a robust measure of pure investment performance.

5.4 Snapshot Distortions

In the current implementation operating in snapshot mode, the absence of real daily prices introduces artificial effects (explicitly visible in diagnostic plots):

- abrupt changes in portfolio values unrelated to market dynamics
- artificial spikes in daily returns

As a result, the computed TWR should be interpreted primarily as a diagnostic indicator rather than as an exact performance metric.

6 IRR vs TWR: Conceptual Comparison

The two metrics implemented in this project serve fundamentally different purposes.

6.1 Money-Weighted vs Time-Weighted Perspective

- **IRR** (money-weighted return) reflects the actual experience of the investor. It takes into account both the timing and the magnitude of external cash flows.
- **TWR** (time-weighted return) isolates the performance of the portfolio itself and removes the effect of external cash flows.

6.2 Sensitivity to Cash Flows

- IRR is highly sensitive to when cash deposits and withdrawals occur.
- TWR is constructed to be invariant with respect to external cash flows.

6.3 Data Requirements

- IRR requires:
 - a complete cash-flow timeline
 - the final portfolio value
- TWR requires:
 - a consistent time series of portfolio values
 - ideally high-frequency (e.g. daily) price data of traded securities

6.4 Interpretation in Practice

- IRR answers the question:

“What was the effective annual return of this specific investment path?”

- TWR answers the question:

“How did the portfolio perform independently of the investor’s decisions?”

6.5 Limitations in the Current Setting

- IRR remains well defined provided that the final portfolio value is correctly specified.
- In the absence of real historical prices, TWR is affected by snapshot distortions and must therefore be interpreted with caution.

This distinction explains why both metrics are implemented: IRR provides a reliable measure of realized performance, while TWR serves as a structurally correct framework prepared for future integration of high-quality price data.

7 Methodological Observations

During the analysis and reconstruction of portfolio performance, several conceptual and methodological issues were identified in commonly available reporting.

7.1 Ambiguity of Performance Metrics

Reported performance metrics are often presented without a clear reference to the underlying methodology. In particular, it may not be specified whether the metric is based on:

- cash flows (e.g. IRR), or
- a time series of portfolio values (e.g. TWR)

This ambiguity complicates both interpretation and independent verification.

7.2 Incomplete Specification of Final Portfolio Value

In some cases, performance formulas are presented without explicit inclusion of the final portfolio value. Since the final value is an essential component of any money-weighted return calculation, such formulations are incomplete and cannot be considered reliable.

7.3 Non-Reproducibility of Reported Results

Due to missing definitions and insufficient transparency of intermediate steps, reported performance values cannot always be reconstructed from available source data. This highlights the importance of exposing intermediate transformations and audit tables.

7.4 Challenges in Data Integration

The absence of a consistent unique identifier and incomplete linkage between datasets further complicate reconstruction of portfolio evolution. This requires explicit reconstruction of both the positions layer and the portfolio value computation layer directly within the analytical pipeline.

8 Audit and Transparency

Intermediate tables are exposed to ensure full traceability of all computations:

- **portfolio_total** – daily total portfolio value in CZK (positions + cash)
- **holdings_total** – value of positions only (excluding cash)
- **cash_multi** – cash balances converted to CZK across currencies
- **cf_external** – aggregated external cash flows (deposits, withdrawals, fee refunds)
- **twr_frame** – daily TWR computation frame including previous value and daily return
- **fx_daily** – daily exchange rates used for normalization
- **fx_raw** – raw exchange-rate history loaded from annual Czech National Bank files before expansion to daily frequency
- **cashflow_fx** – normalized cash-flow table loaded directly from the broker export
- **cashflow_fx** – cash-flow table after linkage with exchange-rate data and conversion to CZK

Each table corresponds to a well-defined transformation step, allowing verification of both data integrity and numerical consistency.

The dimensions of audit tables reported in Appendix A and B are illustrative examples obtained from a specific dataset. They are not fixed constants and may vary depending on file versions, the number of years of exchange-rate history, and the selected evaluation horizon.

9 Visualization

The following plots correspond to the TWR computation for the situation described in Appendix B.1, i.e. the full time range defined by the input dataset.

These visualizations serve as diagnostic tools for understanding portfolio evolution as well as structural limitations of valuation in snapshot mode.

9.1 Portfolio Value

Figure 1 shows the evolution of total portfolio value over time. Abrupt changes indicate snapshot distortions rather than actual market movements.

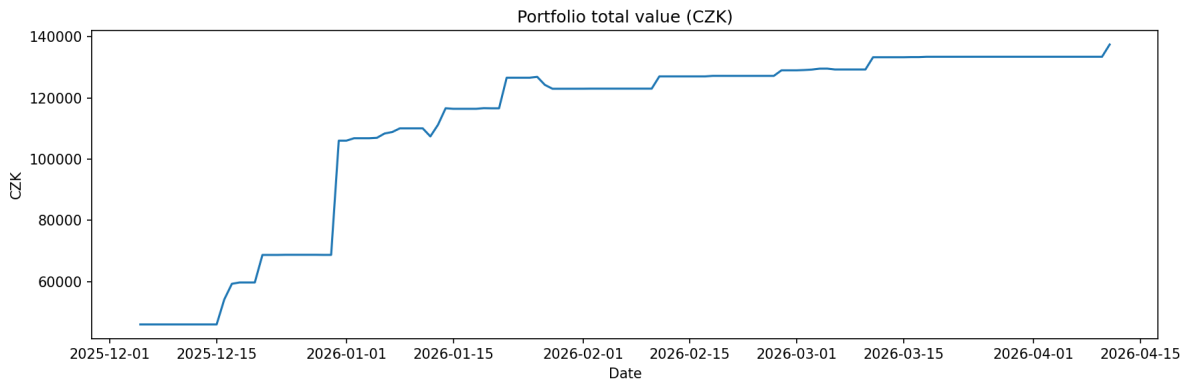


Figure 1: Total portfolio value over time (TWR from Appendix B.1)

9.2 Portfolio Decomposition

Figure 2 decomposes total portfolio value into positions and cash. This highlights transitions between invested and non-invested capital.

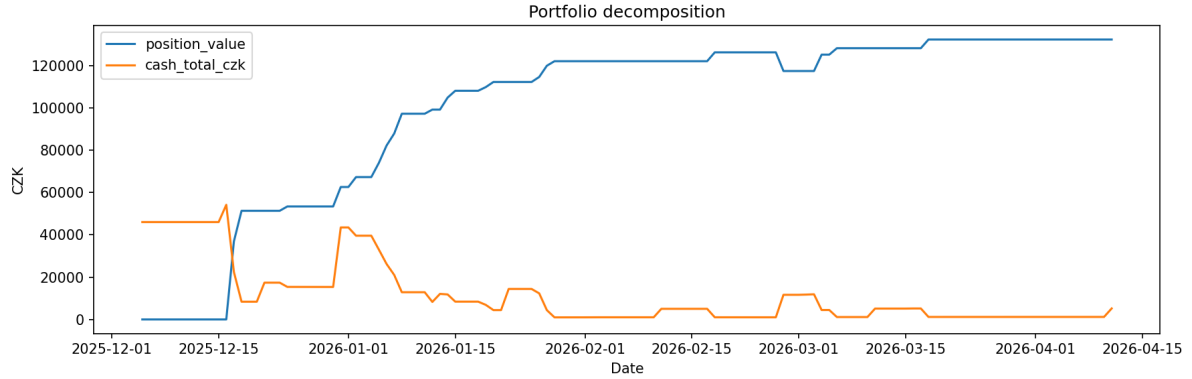


Figure 2: Position value vs cash balance (TWR from Appendix B.1)

9.3 Daily Returns

Figure 3 shows daily portfolio returns. In snapshot mode, artificial spikes reflect inconsistency of the portfolio value time series rather than actual performance.

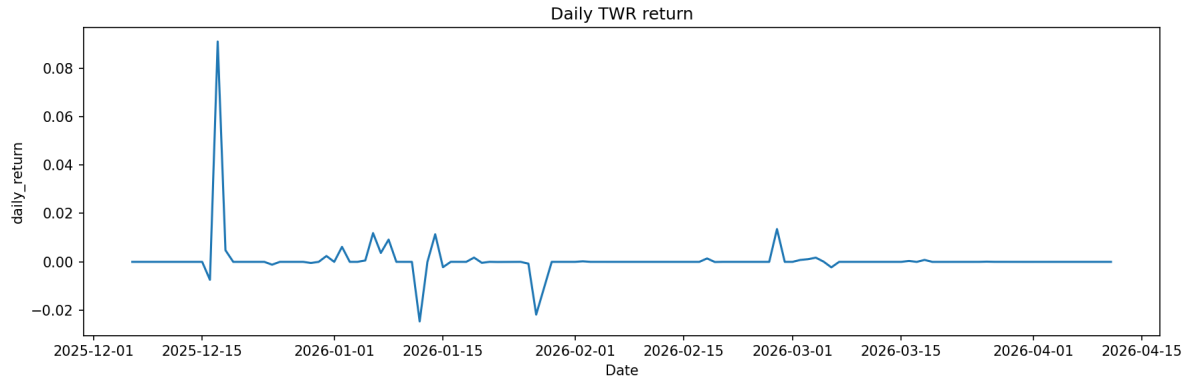


Figure 3: Daily returns with snapshot distortions (TWR from Appendix B.1)

10 Limitations

- absence of real historical prices
- reliance on snapshot mode
- need for manual mapping across datasets
- potential differences in exchange-rate data sources and their usage:
it is not always clear whether identical sources are used, nor which specific fields (e.g. bid, ask, midpoint, or reference rates) are applied in conversion. Even small differences may lead to visible deviations in aggregated results.

11 Conclusion

This project shows that reliable portfolio performance analysis requires a fully explicit and reproducible framework combining clearly defined data transformations with well-specified mathematical models.

Two standard methodologies were implemented:

- IRR as a money-weighted return reflecting the outcome from the perspective of a specific investor, based on a complete and consistently defined cash-flow structure including the final portfolio value
- TWR as a time-weighted return isolating portfolio performance, constructed from a time series of portfolio values and prepared for integration of real historical prices

A central outcome of the project is the identification of structural issues arising when working with real broker data. These include ambiguity in identification of traded securities, incomplete linkage between datasets, and limited transparency in the construction of reported performance metrics. Such issues make independent verification of results impossible without explicit reconstruction of the computational pipeline.

The implemented solution addresses these issues by:

- enforcing a consistent data model across all inputs
- normalizing all monetary quantities through a transparent exchange-rate layer
- separating external cash flows from portfolio value computation
- exposing intermediate results through audit tables

As a consequence, for all computed quantities it is possible to determine retrospectively how they were obtained and from which source data, and to verify this process step by step.

In the current setting, IRR provides a robust and interpretable performance metric because it is based directly on observed cash flows and explicitly specified final portfolio value. In contrast, TWR is affected by the absence of real daily prices, which prevents construction of a fully consistent time series of portfolio values, and therefore serves primarily as a diagnostic framework highlighting structural properties of the data.

The architecture is nevertheless fully prepared for future extension. Once a consistent price history becomes available, the same approach will allow obtaining an economically meaningful time-weighted return without structural changes.

Overall, the project shows that transparency, reproducibility, and precisely defined methodology are essential for meaningful portfolio analysis, especially in environments where source data are incomplete or ambiguously structured.

A Appendix A: IRR Usage

Interpretation

- IRR as a money-weighted return reflects the outcome from the perspective of a specific investor and is sensitive to the timing of cash flows.
- Explicit inclusion of FV_d is essential. Without it, equation (1) does not have a unique solution.

A.1 Run with Two Parameters

Task

Compute IRR for the evaluation date **April 7, 2026**, assuming that the final portfolio value at this date is **129,465.0 CZK**.

Bash

```
python scripts/run_irr_zisk.py --d 2026-04-07 --fv 129465.0
```

Output

```
=== IRR / ZISK RESULT ===
```

```
IrrZiskResult:
  date (d):          2026-04-07
  FV_d:              129,465.00 CZK
  n_cf:              124
  sum_in:            128,610.24 CZK
  sum_cf_net:        128,610.24 CZK

  IRR_d:             0.024043
  IRR_d (p.a.):       2.40 %

  zisk_d:            0.006646
  zisk_d (%):         0.66 %
```

```
=== AUDIT TABLES ===
```

```
cashflow_raw: shape=(235, 7)
fx_raw: shape=(314, 3)
fx_daily: shape=(128, 3)
cashflow_fx: shape=(235, 10)
cashflow_selected: shape=(38, 11)
cashflow_timeline: shape=(128, 2)
```

A.2 Run with One Parameter

Task

Compute IRR using the **current date** as the evaluation date, with the final portfolio value specified as **134,477.6 CZK**.

Bash

```
python scripts/run_irr_zisk_fixed.py --fv 134477.6
```

Output

```
=== IRR / ZISK RESULT ===
```

```
IrrZiskResult:
```

```
date (d):          2026-04-14
FV_d:              134,477.60 CZK
n_cf:              128
sum_in:            132,610.24 CZK
sum_cf_net:        132,610.24 CZK

IRR_d:             0.049513
IRR_d (p.a.):       4.95 %

zisk_d:            0.014082
zisk_d (%):         1.41 %
```

```
=== AUDIT TABLES ===
```

```
cashflow_raw: shape=(235, 7)
fx_raw: shape=(314, 3)
fx_daily: shape=(128, 3)
cashflow_fx: shape=(235, 10)
cashflow_selected: shape=(38, 11)
cashflow_timeline: shape=(128, 2)
```

B Appendix B: TWR Usage

Interpretation

- The computation covers the period from the first available date to the selected final date.
- The result reflects chained portfolio performance over the corresponding dataset.
- In snapshot mode, the result should be interpreted primarily as a diagnostic indicator.

B.1 Default Computation (Full Range)

Task

Compute TWR over the **entire available time range**. (See Figures 1–3.)

Bash

```
python scripts/run_twr_snapshot.py
```

Output

```
=== TWR RESULT ===
```

```
TwrResult:
  start_date:          2025-12-05
  end_date:            2026-04-11
  n_days:              128
  start_value_czk:     46,000.00 CZK
  end_value_czk:       137,439.55 CZK
  total_external_cf_czk: 132,610.24 CZK

  TWR:                 0.089912
  TWR (%):              8.99 %
```

```
=== AUDIT TABLES ===
```

```
portfolio_total: shape=(128, 2)
holdings_total: shape=(128, 2)
cash_multi: shape=(128, 2)
twr_frame: shape=(128, 5)
cf_external: shape=(27, 2)
fx_raw: shape=(314, 3)
fx_daily: shape=(128, 3)
cashflow_raw: shape=(235, 7)
cashflow_fx: shape=(235, 10)
```

B.2 Run with Parameter

Task

Compute TWR only up to the date **December 31, 2025** (shortened analysis period).

Bash

```
python scripts/run_twr_snapshot.py --end-date 2025-12-31
```

Output

```
=== TWR RESULT ===
```

```
TwrResult:
  start_date:          2025-12-05
  end_date:            2025-12-31
  n_days:              27
  start_value_czk:     46,000.00 CZK
  end_value_czk:       104,822.67 CZK
  total_external_cf_czk: 101,103.91 CZK

  TWR:                 0.067676
  TWR (%):             6.77 %
```

```
=== AUDIT TABLES ===
```

```
portfolio_total: shape=(27, 2)
holdings_total: shape=(27, 2)
cash_multi: shape=(27, 2)
twr_frame: shape=(27, 5)
cf_external: shape=(7, 2)
fx_raw: shape=(314, 3)
fx_daily: shape=(128, 3)
cashflow_raw: shape=(235, 7)
cashflow_fx: shape=(235, 10)
```